Blindspot

Your dashboard shows what happened. BlindSpot shows what it’s hiding.

The analyst a small shop can’t afford, built so every number proves itself.

Live v2 · The proof chain · Claude, fact-checked

blindspot-pro.onrender.com



THE PROBLEM, IN NUMBERS

36.2M

small businesses in the US: 45.9% of private-sector workers.

SBA Office of Advocacy, Feb 2026

53%

of small-business leaders don't know what insights their own data could give them.

AWS survey of 800+ SMB leaders, 2023 (vendor survey)

29.0% vs 7.7%

unprofitable: little repeat revenue vs over half from repeat customers.

Small Business Expo survey of 1,100+ owners, 2026. An association, not proof of cause.

The data to catch problems already exists. Nobody reads it.

THE DASHBOARD LIES

A real shop. Its dashboard said:

Sales vs the same weeks last year

+6.2%

Looks healthy. Everyone relaxes.

UCI Online Retail II: a real UK online shop's public sales history (25% customer sample).

What was hiding underneath:

- **10** — best customers stopped ordering · £93,531 a year
- **51% → 42%** — new customers coming back (tested, p = 0.002) · fewer second orders
- **23** — products that sold last year now sell nothing · £36,302
- **509** — order lines reversed within 2 days · £114,284 padding the total
- **7,872** — duplicate lines in the file · counted once by BlindSpot

WHAT BLINDSPOT IS

Drop in your sales export. BlindSpot finds what the totals hide, proves it on the exact rows, and tells you what to do.

Read
Any export, PDF or photo of a table

Clean
Duplicates, reversals, returns, fees

Prove
Statistics, exact rows, past replays

Act
Contact, then recheck who came back

For anyone who sells to repeat customers: online shops, wholesalers, makers. No analyst needed.

EVERYTHING IN THE APP

12 tools. One connected workspace.

Overview
Health score, the fix-first problem, a checked AI briefing

Data Sources
Shopify, CSV, Excel, PDF, photo

Source Data
Every row, and how it was cleaned

AI Analyst
Ask in plain English, see the tools used

Risk Intelligence
Problems ranked by money, labelled by certainty

Customer Radar
Who is slipping, and the follow-up loop

Stock & Cash
Sales gaps that look like stock-outs

Prove It
Did the promo really work?

Simulator
What a fix could be worth

Forecast
Next months, with its past accuracy

Dashboard Audit
Misleading charts, color-blind checks

Reports
30/90/180-day plan, 30 currencies

★ THE GAME CHANGER

The proof chain.

Dashboards give you a number and ask you to trust it. CRMs list customers but never tell you who is slipping or whether it's real. **BlindSpot makes every number prove itself, then shows whether acting on it paid off.**

Finding
10 quiet customers, £93,531

→

Exact rows
add up to the penny

→

Tested on the past
flagged stayed away far more

→

Act
checked AI note, you send it

→

Recheck
who ordered after your contact

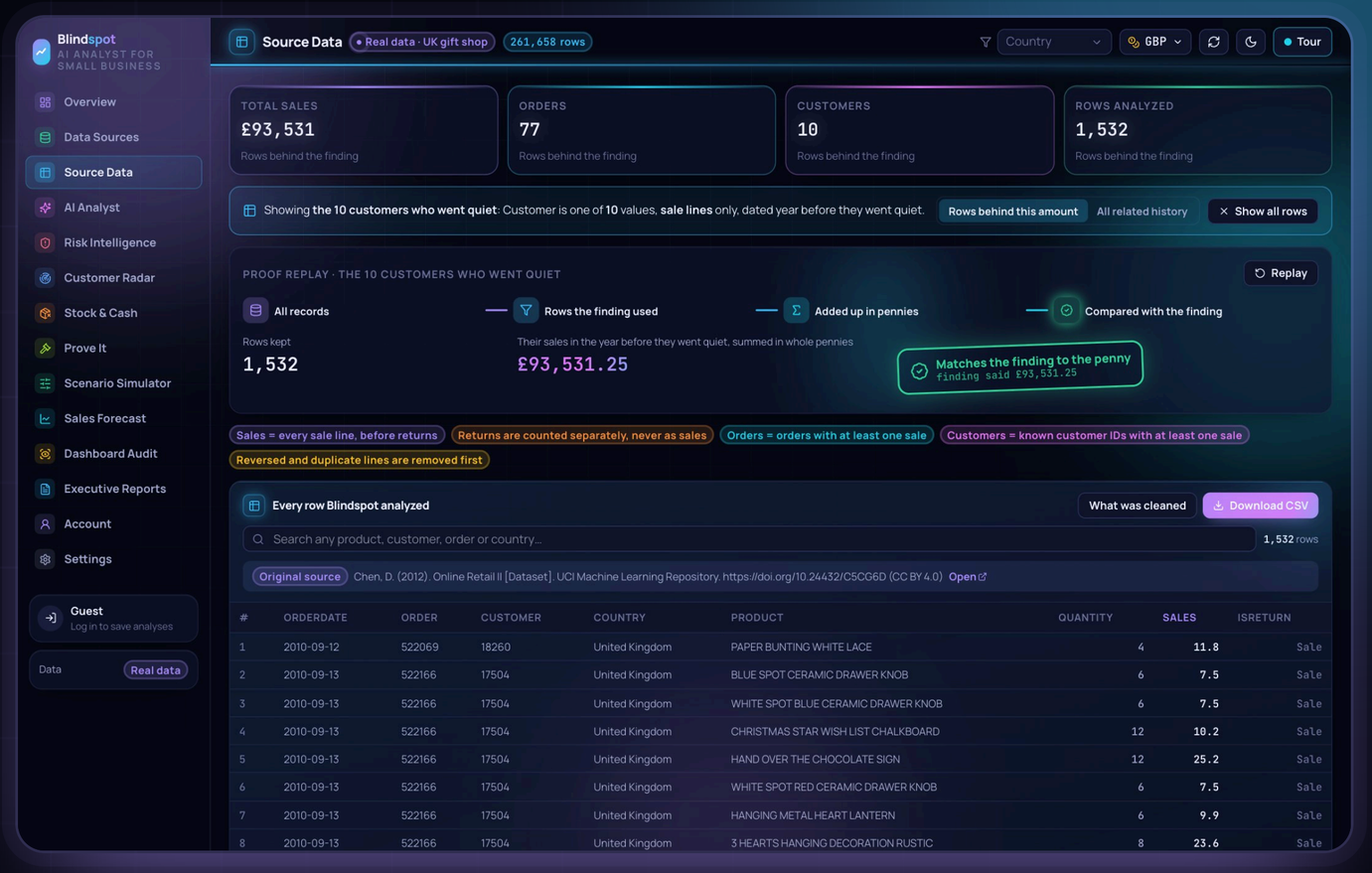
Finding → proof → action → result. No dashboard does that loop.

PROOF CHAIN · STEP 2

Every number opens into its rows.

```
Finding £93,531.25
Rows behind it £93,531.25 ✓
```

A separate Python script, sharing no code with the app, recomputes the headline numbers: **7 of 7 match**, and all **10 accounts** match row for row (orders, usual gap, silence, spend).



PROOF CHAIN · STEPS 3–5

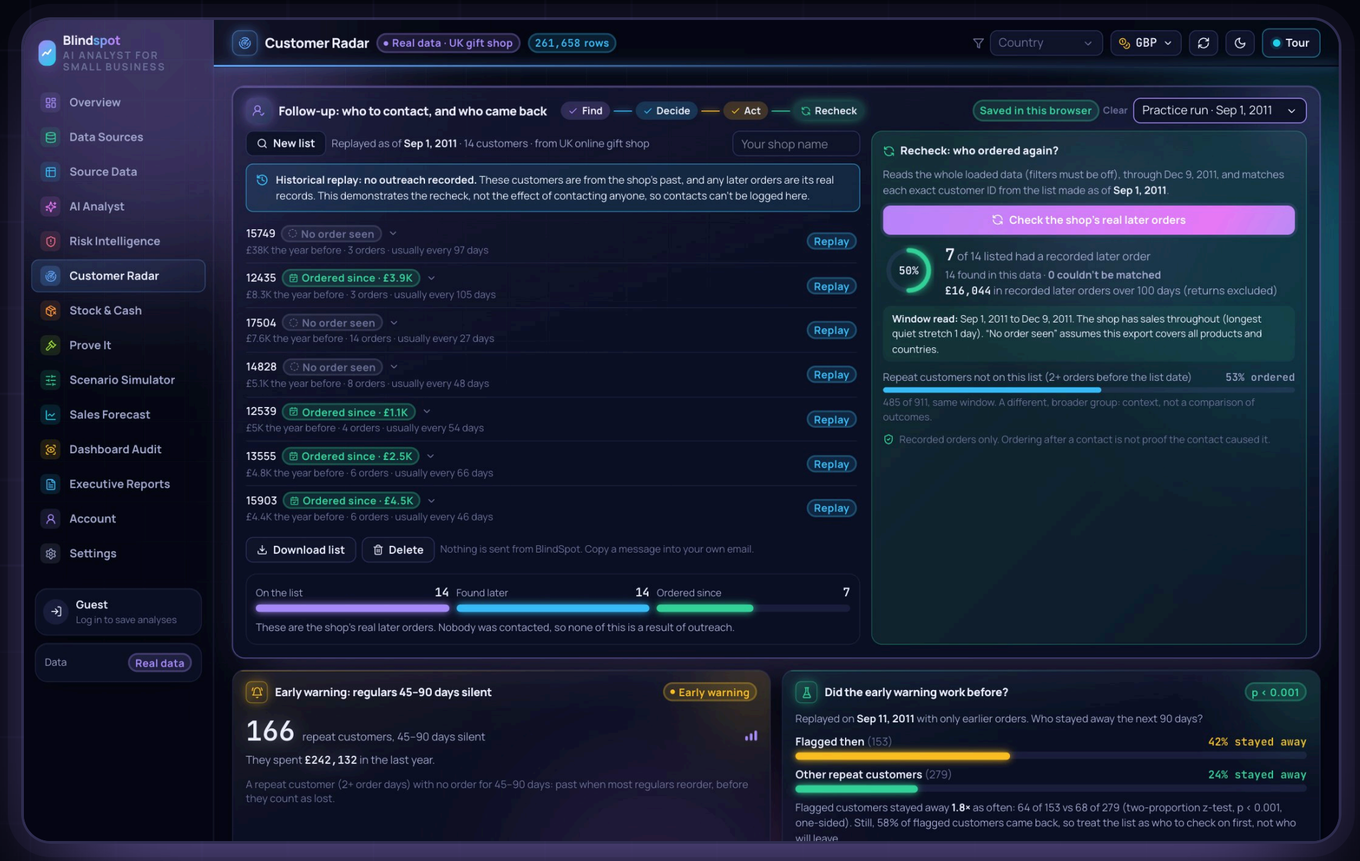
Tested on the past. Rechecked after you act.

The rule, replayed on the shop's own history (180 days later)

32–36% **3–8%**

flagged vs other top customers who stayed away · Fisher exact p < 0.005 · most still came back

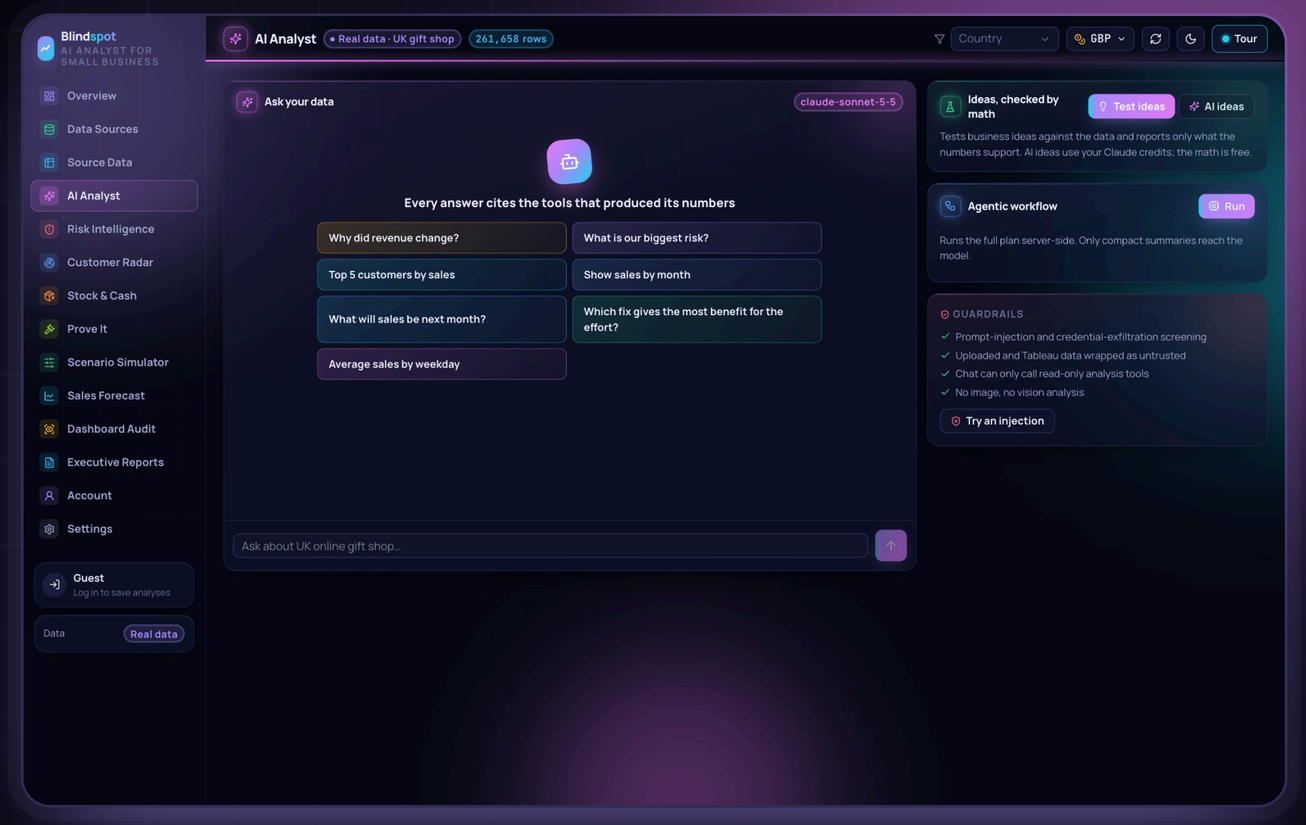
Recheck counts only orders after each customer's own contact day. Historical replay, no outreach recorded: 7 of 14 ordered again.



AI, ON A LEASH

Claude explains. The math decides.

- 41 Zod-typed tools over REST + Model Context Protocol; Claude gets 21 read-only ones
- A fact-checker parses every number Claude writes and matches it to tool output; misses go back once
- Outreach notes rejected for amounts, dates, discounts or “you left”; products filled from purchase history
- Structured outputs (JSON schema), agent loop, usage ledger, 12/min rate limit



UNDER THE HOOD

Six layers, from your file to a checked answer.

1

Reader · in your browser

SheetJS | Papa Parse + own separator detection | pdf.js word x/y | 8 languages | Claude vision → editable table

2

Server · Node + Express

isolated sessions | 512 MB budget | dictionary-encoded 1M rows | caches warmed before listening | WebSocket bridge

3

Engine · pure TypeScript, zero AI

deterministic | memoized per dataset | same-weeks-last-year windows

4

Proof

drill-through to the penny | independent Python check | replays on the past

5

AI layer · Claude Sonnet 5.5

41 typed tools | MCP | fact-checker | draft guard

6

Accounts · Supabase Postgres

PKCE sign-in | row-level security | tested with throwaway users

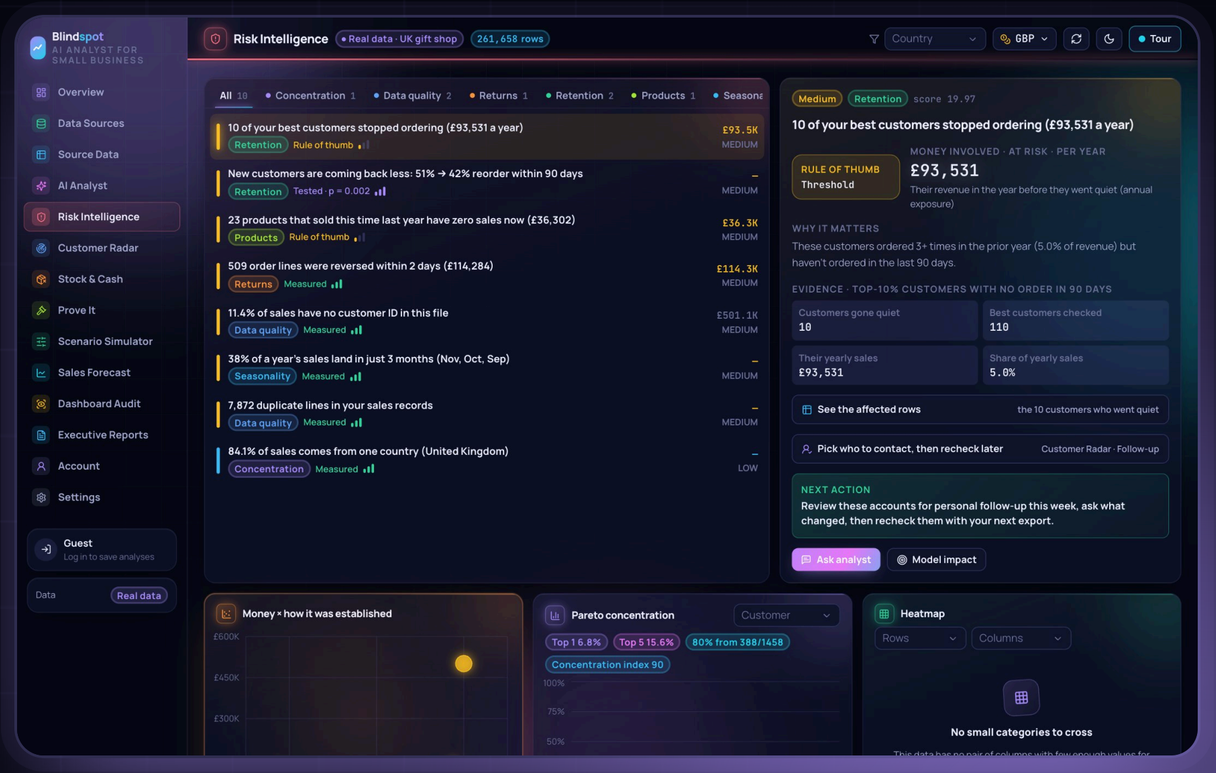
THE STATISTICS ENGINE

Written from scratch. Honest about doubt.

Welch t-test · two-proportion z-test · Fisher exact · bootstrap ×2,000 · Benjamini-Hochberg FDR · weekly revenue share · same weeks last year

Every finding is labelled

Measured · Tested · p-value · Rule of thumb · Possible, not confirmed · Not judged



It downgraded its own most dramatic finding, a best-seller "down 40%", to **"possible, not confirmed" (q = 0.27)** once orders were treated as dependent.

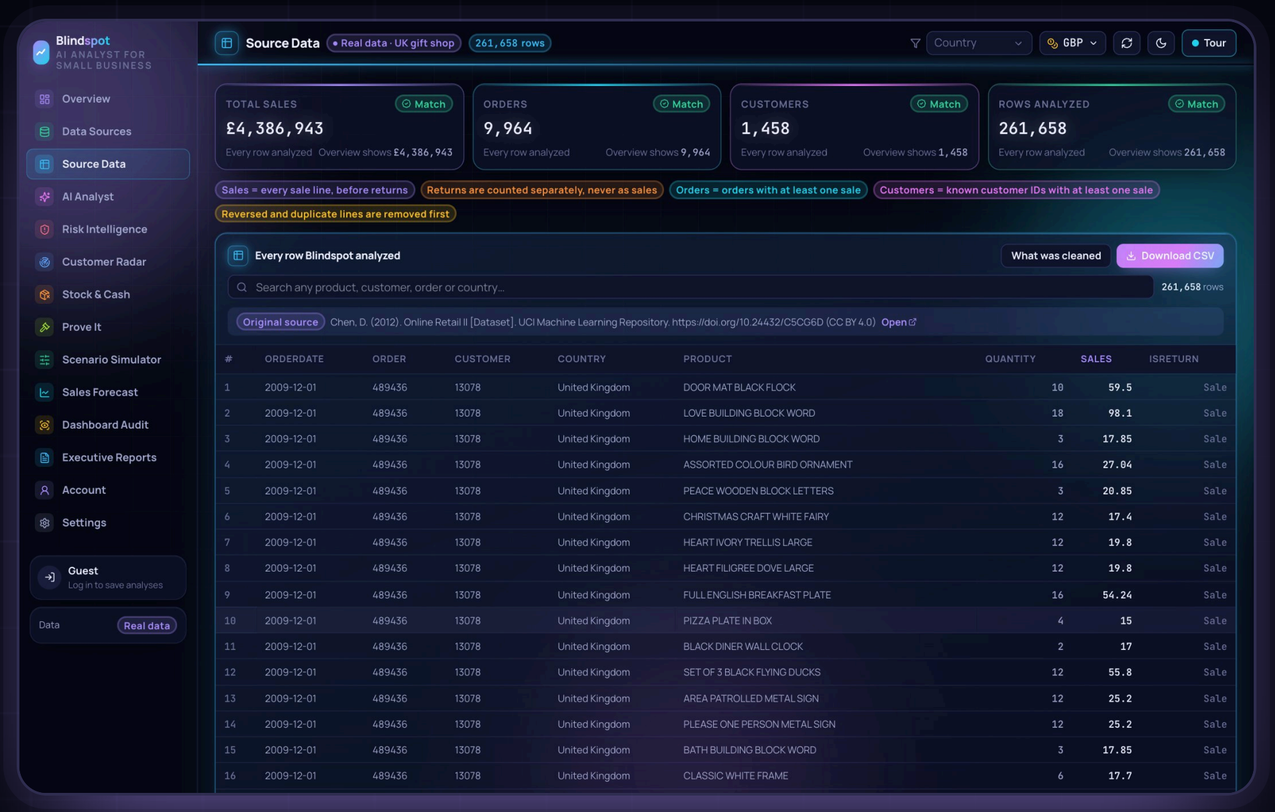
THE READER

Messy files in. Clean records out.

Shopify, Etsy, Amazon, Square, WooCommerce, Stripe exports · CSV, Excel, PDF reports · a photo of a table (reviewed before use).

- PDFs store words, not tables: rows rebuilt from each word's position
- columns scored by name *and* values, in 8 languages
- cancelled orders paired with the sale they reverse
- junk files (notes, JSON, web pages) get a plain answer, never a crash

1,067,371 real rows cleaned in ~40 s and analysed in ~6 s on a 2020 laptop.



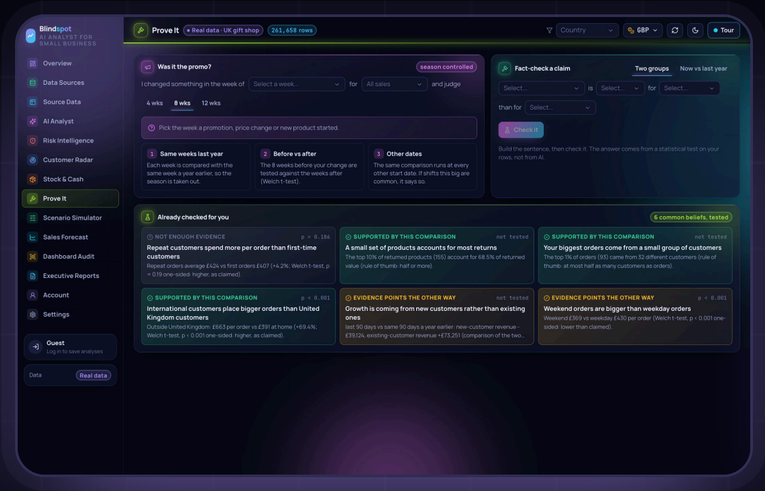
BEYOND CUSTOMERS

Forecasts that show their track record. Promos put on trial.



Sales Forecast

Replayed on 12 past months: 9.1% average miss, shown next to every forecast



Prove It

Was it the promo? Compared with the same weeks last year and every other start date



Dashboard Audit

Grades charts for misleading axes and colors color-blind viewers can't tell apart

PRIVATE AND SAFE BY DESIGN

Your customers’ data stays yours.

Row-level security

User A can’t read, change or forge user B’s lists. Tested with throwaway users.

Read-only access

Shopify connects read-only. Keys never stored. Nothing is ever sent for you.

Guest mode

Try everything without an account; private sessions per visitor.

Honest labels

“Ordered after contact” is never called proof the contact worked.

No fake numbers

Every AI number checked; every claim tagged with how sure it is.

Open evidence

VALIDATION.md and scripts let anyone reproduce the key numbers.

BY THE NUMBERS

Built to hold up under scrutiny.

~40,000
lines of TypeScript, Python and SQL

290+
automated tests, plus 6 browser tests

41
typed tools · 21 for Claude

1,067,371
real sales rows tested

7 / 7
headline numbers match an independent check

30
currencies at today's rates

Blindspot

Stop trusting totals.
Start seeing what they hide.

blindspot-pro.onrender.com

github.com/Sidd-solves/BlindSpot

Next: a real shop owner on their own export, live connectors for WooCommerce and Square, weekly alerts.